

Two Short Proofs of a Bounded-Valuation Theorem in Multiplicative Persistence

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Abstract

Fix a base $b \geq 2$, a prime $p \mid b$, and a finite multiset of digits different from 1, while allowing arbitrarily many additional digits equal to 1 and no zero digits. We give two short proofs that the p -adic valuations of the resulting integers are bounded. The first uses a finitely branching tree of divisible suffixes and König's lemma; the second is an equivalent p -adic compactness argument obtained by stabilizing digits from right to left. In base ten with $p = 2$, this gives a qualitative proof of Conjecture 2 of Brier, Clavier, Gutsche, and Naccache. Fonga gives a quantitative proof of the conjecture, obtaining explicit bounds and an effective finite procedure for the even-target persistence program.

1 Statement and context

Let $b \geq 2$. Fix nonnegative integers m_d for $d \in \{2, \dots, b-1\}$. Let $\mathcal{A}$ be the family of positive integers whose base- b expansions contain exactly m_d copies of digit d for each $d \geq 2$, contain no zero digit, and contain an arbitrary number of copies of digit 1.

Theorem 1. *Let p be a prime divisor of b . Then*

$$\{v_p(n) : n \in \mathcal{A}\}$$

is bounded.

For $b = 2$, the set of possible digits $2, \dots, b-1$ is empty, so the fixed multiset is necessarily empty. Thus $\mathcal{A}$ consists only of binary repunits $2^r - 1$, all of which are odd. Hence Theorem 1 is immediate in base 2. We therefore assume $b \geq 3$ in the substantive arguments below.

In decimal notation and for $p = 2$, Theorem 1 is precisely the qualitative boundedness assertion in Conjecture 2 of Brier, Clavier, Gutsche, and Naccache [1]. Fonga gives a quantitative proof of the conjecture with explicit bounds depending on the prescribed digit multiplicities and gives an effective finite method for the even-target persistence program [2]. The arguments below are instead short qualitative proofs based on compactness.

2 Suffix-tree proof

Assume for contradiction that v_p is unbounded on $\mathcal{A}$.

For each integer $e \geq 1$, choose $x \in \mathcal{A}$ satisfying

$$v_p(x) \geq \lceil (e-1) \log_p b \rceil + 1.$$

Then $x \geq p^{v_p(x)} > b^{e-1}$, so x has at least e base- b digits. Moreover, since $p \mid b$, we have $\log_p b \geq 1$, so the displayed lower bound is at least e . Hence $p^e \mid x$.

Let y_e be the suffix formed by the final e base- b digits of x . Because $y_e \equiv x \pmod{b^e}$ and $p^e \mid b^e$, we have $p^e \mid y_e$.

Construct a rooted tree whose vertices at depth e are the e -digit, zero-free base- b strings that use each digit $d \geq 2$ at most m_d times and whose numerical values are divisible by p^e . The parent of a depth- $(e+1)$ vertex is obtained by deleting its most significant digit.

Parent closure is valid: if

$$y_{e+1} = a_e b^e + y_e$$

and $p^{e+1} \mid y_{e+1}$, then $p^e \mid a_e b^e$ because $p \mid b$, and therefore $p^e \mid y_e$. After adjoining an empty root, the tree is finitely branching and has a vertex at every depth. König's lemma therefore gives an infinite branch.

Along this branch only finitely many digits can differ from 1. Choose N beyond the final non-1 digit, and let A_N be the integer represented by the first N branch digits, counted from the least significant end. For every $e > N$,

$$y_e = A_N + \sum_{j=N}^{e-1} b^j = A_N + \frac{b^e - b^N}{b - 1}.$$

Since $p^e \mid y_e$ and $p^e \mid b^e$,

$$p^e \mid (b - 1)A_N - b^N$$

for arbitrarily large e . A fixed nonzero integer cannot be divisible by arbitrarily high powers of p , so

$$(b - 1)A_N = b^N.$$

Reducing modulo $b - 1$ gives $0 \equiv 1 \pmod{b - 1}$, impossible because $b \geq 3$. This proves Theorem 1. $\square$

3 Equivalent p -adic compactness proof

Suppose again that the valuations are unbounded. Choose $x_k \in \mathcal{A}$ with $v_p(x_k) \rightarrow \infty$. Their base- b lengths tend to infinity because $x_k \geq p^{v_p(x_k)}$, and $x_k \rightarrow 0$ in $\mathbb{Q}_p$.

Read digits from right to left. By diagonal subsequence selection, pass to a subsequence for which each fixed digit position eventually stabilizes. Write the limiting digits as $a_0, a_1, \dots$. The fixed budget of non-1 digits implies that $a_j = 1$ for all sufficiently large j . Since $p \mid b$, the series

$$L = \sum_{j=0}^{\infty} a_j b^j$$

converges in $\mathbb{Q}_p$.

To justify convergence of the selected integers to L , fix $r \geq 1$ and choose J with $p^r \mid b^J$. For all sufficiently large k , the first J digits of x_k agree with $a_0, \dots, a_{J-1}$ and the length of x_k exceeds J . Thus

$$x_k - L \in b^J \mathbb{Z}_p \subseteq p^r \mathbb{Z}_p.$$

Hence $x_k \rightarrow L$. Since also $x_k \rightarrow 0$, uniqueness of limits gives $L = 0$.

For N beyond the last non-1 limiting digit,

$$0 = L = A_N + \sum_{j=N}^{\infty} b^j = A_N - \frac{b^N}{b-1}.$$

Here $p \nmid b-1$, so $b-1$ is a unit in $\mathbb{Z}_p$ and the geometric-series identity is valid p -adically. Thus $(b-1)A_N = b^N$, contradicting reduction modulo $b-1$ as above.

4 Scope

The arguments above are qualitative: they prove that a bound exists but do not produce a computable optimal bound from the prescribed digit multiplicities. Consequently, by themselves they do not supply the effective cutoff required to execute the backward-search program for nonzero even terminal digits in [1]. Fonga’s quantitative theorem supplies that missing ingredient [2].

The result also does not prove that decimal multiplicative persistence is at most 11. It establishes only the fixed-digit-multiset boundedness statement.

Repository

Code, certificates, and reproducibility materials accompanying this note are available at <https://github.com/00028ECB/multiplicative-persistence-bounded-valuation>.

References

- [1] E. Brier, C. Clavier, L. Gutsche, and D. Naccache, *The Multiplicative Persistence Conjecture Is True for Odd Targets*, arXiv:2110.04263, 2021.
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